Week 7 — Quantum States of Light: The DV and CV Languages

EEQT30001 Quantum Technology and Engineering, Unit 2 (Quantum Optics and Quantum Sensing)

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Learning objectives (Week 7)

1. Explain why each electromagnetic mode is quantized as a harmonic oscillator, and use $[a, a^\dagger] = 1$ to compute properties of Fock, coherent, and squeezed states.
2. Draw vacuum, coherent, and squeezed states in phase space, and state the two encodings of light-based qubits (DV vs. CV).
3. Classify a light source from $g^{(2)}(0)$ (thermal 2 / coherent 1 / single photon 0) and explain why antibunching — not the photoelectric effect — proves the photon.
4. Describe how a cavity redesigns the vacuum (Jaynes–Cummings physics, the Purcell effect) and how an SNSPD detects a single photon.

Reading: Fox, *Quantum Optics*, Ch. 5–7 (Secs. 1–2), Ch. 10 (Sec. 3). Full derivations: QO_note_2, 3, 5, and 4.

Opening puzzle. Textbooks say the photoelectric effect proved that light is made of photons. It did not: a *semiclassical* theory — quantized atoms driven by a purely classical field — reproduces every feature of the photoelectric effect, including the frequency threshold. The experiment that actually certified the photon came seventy years after Einstein: photon *antibunching*, observed in 1977 [1]. This week builds the language needed to appreciate how much work it takes to prove that light arrives "one at a time", and what non-classical light is good for.

1 Quantum States of Light: The DV and CV Languages

1.1 One mode is a harmonic oscillator

Maxwell's equations in a source-free region are linear, so their solutions organize into *eigenmodes*: field patterns that oscillate at a single frequency ω, with a shape fixed by the geometry (a plane wave in free space, a standing wave in a cavity, a guided mode in a waveguide). The energy of one mode is quadratic in the field amplitudes $(\mathbf{E}, \mathbf{B})$ — formally identical to the energy $\frac{p^2}{2m} + \frac{m\omega^2 x^2}{2}$ of a mechanical oscillator. Canonical quantization therefore proceeds exactly as for a particle in a parabolic well: promote the canonical pair to operators, impose the commutator, and change variables to ladder operators (QO_note_2, Sec. 1–2). The result, for every mode,

$$\mathcal{H} = \hbar\omega\left(a^\dagger a + \tfrac{1}{2}\right), \qquad [a, a^\dagger] = 1. \tag{1.1}$$

A *photon* is one quantum of excitation of a mode — the n-th rung of that mode's energy ladder — not a small ball flying through space. **The mode comes first; the photon is the mode's excitation.** Everything in this unit follows from the single algebraic statement $[a, a^\dagger] = 1$.

1.2 The vacuum is not empty

The ground state $|0\rangle$ of Eq. (1.1) has energy $\hbar\omega/2$ and, more importantly, ***fluctuating fields***: $\langle 0|\mathbf{E}|0\rangle = 0$ but $\langle 0|\mathbf{E}^2|0\rangle \neq 0$. These zero-point fluctuations are physical. Summing $\hbar\omega/2$ over the modes allowed between two parallel mirrors, and comparing with free space, yields a measurable attraction (Casimir, 1948; full derivation in QO_note_2, Example 4),

$$\frac{F}{A} = -\frac{\pi^2 \hbar c}{240\, d^4}. \tag{1.2}$$

The lesson to remember for Sec. 3 and for Week 10: the vacuum has *structure* — which modes exist, and how strong their zero-point fields are at a given point, depends on the boundary conditions. Structure can be engineered.

1.3 Fock states: the DV language

The eigenstates $|n\rangle$ of $a^\dagger a$ (*Fock* or number states) are maximally quantum: the field expectation value $\langle n|\mathbf{E}|n\rangle$ vanishes for every n, so a Fock state carries ***no phase*** — only fluctuations (QO_note_2, Sec. 2.3). Encoding information in discrete photonic degrees of freedom — photon number ($|0\rangle, |1\rangle$), polarization ($|H\rangle, |V\rangle$), or path — is called *discrete-variable* (DV) encoding. It is conceptually clean but experimentally expensive: single photons are hard to make and easy to lose.

1.4 Quadratures and phase space: the CV language

The Hermitian combinations of a and $a^\dagger$,

$$X = \frac{a + a^\dagger}{\sqrt{2}}, \qquad P = \frac{a - a^\dagger}{i\sqrt{2}}, \qquad [X, P] = i \;\Rightarrow\; \Delta X\, \Delta P \geq \tfrac{1}{2}, \tag{1.3}$$

are the dimensionless position and momentum of the mode — the *quadratures*.[1] They are continuously valued and directly measurable (by homodyne detection, Week 10), and encoding information in them is called *continuous-variable* (CV) encoding.

Phase space gives the picture. A classical oscillator is a ***point*** (x, p) tracing a circle as it evolves. A quantum state cannot be a point — Eq. (1.3) forbids sharp X and P simultaneously — so even the "most classical" quantum state is a fuzzy blob of area $\sim \frac{1}{2}$, rotating rigidly at ω (Fig. 1).

1.5 Coherent states: the most classical light

Which quantum state behaves like the classical field $E_0 \cos(\omega t + \phi)$ of a laser? Demanding $\langle \mathbf{E} \rangle \propto$ the classical field leads to the eigenvalue problem (QO_note_3, Sec. 2)

$$a|\alpha\rangle = \alpha|\alpha\rangle, \qquad |\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \tag{1.4}$$

with α a complex number — the dimensionless amplitude ***and*** phase of the wave. Key properties:

[1] The original lecture notes QO_note_2–5 use the convention $X_s = (a + a^\dagger)/2 = X/\sqrt{2}$, $Y_s = P/\sqrt{2}$, for which $[X_s, Y_s] = i/2$ and $\sigma(X_s)\sigma(Y_s) \geq 1/4$. All statements convert by a factor $\sqrt{2}$ per quadrature.

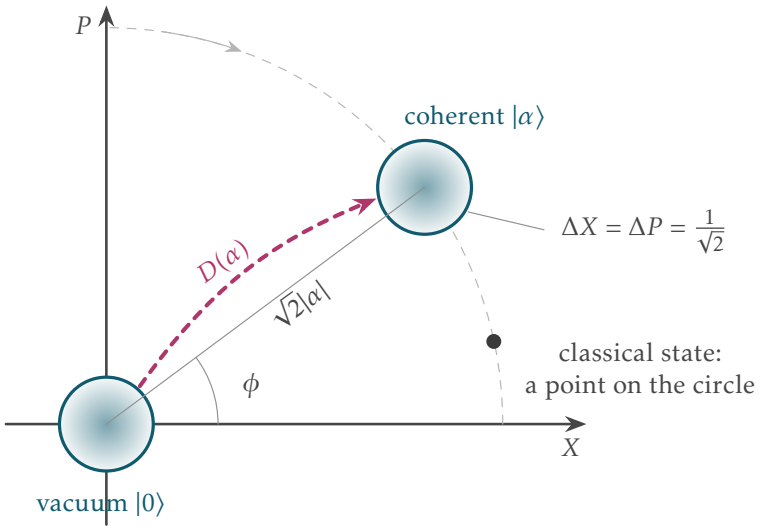


Figure 1: Phase space in our convention, Eq. (1.3). A classical state is a point tracing a circle; a quantum state is a minimum-uncertainty blob of area $\sim \frac{1}{2}$. The vacuum sits at the origin; the displacement $D(\alpha)$ shifts it to $(\langle X\rangle, \langle P\rangle) = \sqrt{2}\,(\mathrm{Re}\,\alpha, \mathrm{Im}\,\alpha)$, a distance $\sqrt{2}|\alpha|$ from the origin, without changing its shape — a coherent state, $\Delta X = \Delta P = 1/\sqrt{2}$.

- Photon number is Poisson-distributed, $p_n = e^{-\langle n\rangle}\langle n\rangle^n/n!$, with $\langle n\rangle = |\alpha|^2$ and $\Delta n = \sqrt{\langle n\rangle}$ — photons in a laser beam arrive ***randomly and independently***, like raindrops.
- It is a minimum-uncertainty state, $\Delta X = \Delta P = 1/\sqrt{2}$: the same circular blob as the vacuum, displaced to $\sqrt{2}\,\alpha$ in the X–P plane. Indeed $|\alpha\rangle = D(\alpha)|0\rangle$ with the displacement operator $D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a)$ — and $D(\alpha)$ is exactly the evolution generated by a classical oscillating current (an antenna, a laser drive; QO_note_4, Sec. 3). A laser ***displaces the vacuum.***
- Free evolution just rotates the blob: $\alpha(t) = \alpha(0)e^{-i\omega t}$.

1.6 Squeezed states: reshaping the vacuum blob

The uncertainty product bounds the *area* of the blob, not its shape. The squeeze operator (QO_note_5, Sec. 2)

$$S(\xi) = \exp\!\left(\tfrac{1}{2}\xi^* a^2 - \tfrac{1}{2}\xi\,(a^\dagger)^2\right), \qquad \xi = re^{i\theta}, \tag{1.5}$$

turns the circle into an ellipse: for $\theta = 0$, $\Delta X = e^{-r}/\sqrt{2}$ and $\Delta P = e^{+r}/\sqrt{2}$ — one quadrature quieter than the vacuum itself, at the price of the other (Fig. 2). Because $S(\xi)$ is quadratic in $a^\dagger$, generating squeezed light requires a ***nonlinear*** process ($\chi^{(2)}$ parametric down-conversion, Week 8). Why care? Metrology: LIGO injects squeezed vacuum to measure below the vacuum noise level (Week 10).

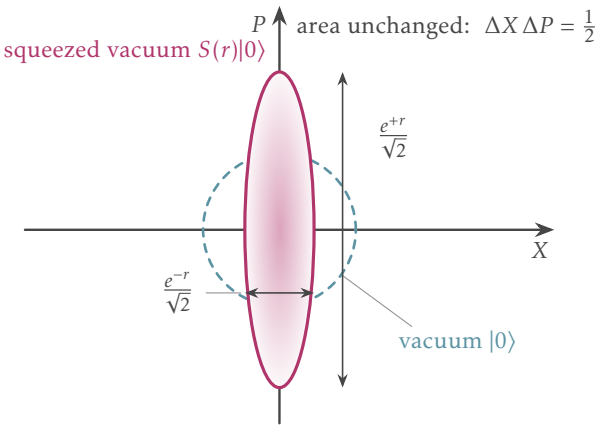


Figure 2: Squeezed vacuum in phase space: one quadrature narrows to $e^{-r}/\sqrt{2}$ at the price of the conjugate one; the area $\Delta X\,\Delta P = \frac{1}{2}$ is unchanged (dashed: the vacuum circle).

1.7 DV vs. CV at a glance

	DV (discrete)	**CV (continuous)**
Basic objects	Fock states, polarization	quadratures X, P
Qubit example	$\|H\rangle, \|V\rangle$; dual-rail $\|01\rangle, \|10\rangle$	coherent/squeezed states; cat, GKP codes
Measurement	single-photon detectors	homodyne/heterodyne
Strength	clean logic, long-distance QKD	deterministic sources, scalable
Weakness	probabilistic sources, loss	noise accumulates

Hybrid schemes (converting DV qubits into cat-state qubits and back) are an active research direction (QO_note_3, introduction). This DV/CV pair of languages is the backbone of the whole unit: Week 8 speaks mostly DV, Week 9 uses both, Week 10 mostly CV.

Summary: Section 1

Every mode of light is a harmonic oscillator: $[a, a^\dagger] = 1$, $\mathcal{H} = \hbar\omega(a^\dagger a + \frac{1}{2})$. The vacuum is a real physical state with fluctuating fields (Casimir). Fock states have sharp n and no phase (DV); coherent states are displaced vacua with Poisson statistics, $\Delta n = \sqrt{\langle n\rangle}$ (the most classical light); squeezed states beat the vacuum noise in one quadrature (CV). Phase space: point (classical) $\to$ blob of area $\sim \frac{1}{2}$ (quantum).

2 Proving the Photon: The Second-Order Correlation $g^{(2)}(0)$

2.1 Photon-counting statistics

Modern detectors do not record a continuous intensity; they *click*. The statistics of clicks classify light (QO_photon_stat):

- **Poissonian**, $\Delta n = \sqrt{\langle n\rangle}$: coherent light (laser).
- **Super-Poissonian**, $\Delta n > \sqrt{\langle n\rangle}$: thermal/chaotic light.
- **Sub-Poissonian**, $\Delta n < \sqrt{\langle n\rangle}$: *no classical explanation* — non-classical light.

A practical warning: loss randomizes. If photons survive a lossy channel or an inefficient detector independently with probability η, sub-Poissonian statistics degrade toward Poissonian — observing quantum statistics requires high collection and detection efficiency. This motivates the detectors of Sec. 3.3.

2.2 The HBT experiment and $g^{(2)}$

Counting statistics could in principle do the job, but Sec. 2.1 showed that loss drags them toward Poissonian. What survives loss is a *ratio*: under linear loss η both $\langle a^\dagger a^\dagger a a\rangle$ and $\langle a^\dagger a\rangle^2$ scale as η^2, so $g^{(2)}(0)$ is unchanged — a 10%-efficient detector measures the same $g^{(2)}(0)$ as a perfect one. That is why the *intensity–intensity* correlation, measured in the Hanbury Brown–Twiss (HBT) geometry, became the standard: split the beam 50:50 onto two detectors and count coincidences (Fig. 3). For a single mode, with the detector-ordering of operators from Glauber's photodetection theory (QO_note_5, Sec. 1),

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger a a\rangle}{\langle a^\dagger a\rangle^2} = 1 + \frac{\sigma^2(n) - \langle n\rangle}{\langle n\rangle^2}. \tag{2.1}$$

$$g^{(2)}(0) = \begin{cases} 2 & \text{thermal light (bunching),} \\ 1 & \text{coherent light (random arrivals),} \\ \dfrac{n-1}{n} \xrightarrow{n=1} 0 & \text{Fock state (antibunching).} \end{cases} \tag{2.2}$$

For *any* classical intensity distribution, the Cauchy–Schwarz inequality forces $g^{(2)}(0) \geq 1$: classical waves can bunch, never antibunch. Therefore a measured $g^{(2)}(0) < 1$ has no classical explanation, and $g^{(2)}(0) \to 0$ certifies a single-photon source. This is the experiment that finally proved the photon: resonance fluorescence from a dilute sodium atomic beam (fewer than one atom in view on average), Kimble, Dagenais and Mandel (1977) [1] — seventy years after the photoelectric effect. The photoelectric effect needs only quantized *atoms*; antibunching needs quantized *light*.

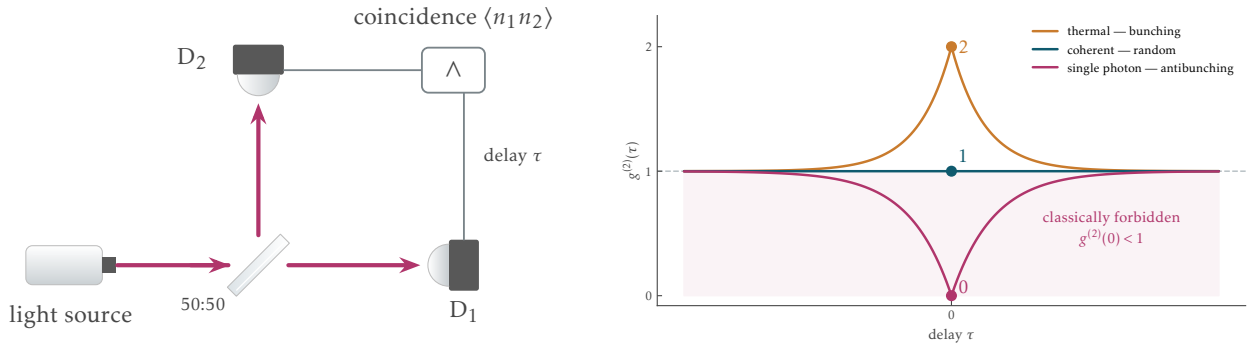


Figure 3: Left: HBT setup — beam splitter, two detectors, coincidence counter. Right: $g^{(2)}(\tau)$ for thermal, coherent, and single-photon light; the shaded region $g^{(2)}(0) < 1$ is classically forbidden.

In-class discussion 1. *A laser is attenuated to* 0.1 *photons per pulse on average. Is it a single-photon source?* — No. Attenuation preserves coherent-state statistics: $g^{(2)}(0) = 1$, and the two-photon probability $p_2 \approx \langle n\rangle^2/2$ is nonzero. Dim is not quantum.

In-class discussion 2. *You measure* $g^{(2)}(0) = 0.5$. *What is the source?* — Non-classical (below 1), but not a perfect single-photon source: e.g. two independent single-photon emitters ($g^{(2)} = 1/2$ exactly), or one imperfect emitter with background.

2.3 Try it in three lines (QuTiP)

QuTiP demo: $g^{(2)}(0)$ of Fock, coherent, and squeezed states

```
from qutip import *
N = 30
states = {"Fock |1>": basis(N,1), "Coherent": coherent(N,2.0),
          "Squeezed": squeeze(N,1.0)*basis(N,0)}
a = destroy(N)
for name, psi in states.items():
    g2 = expect(a.dag()*a.dag()*a*a, psi)/expect(a.dag()*a, psi)**2
    print(name, "g2(0) =", round(g2,3))
```

This is the seed of the bonus homework (published after this week; Colab recommended; worth up to +5 points on Midterm 2).

2.4 Single-photon sources compared

Source	$g^{(2)}(0)$ (typ.)	Rate	Operating T	Deterministic?
Attenuated laser	1 (always)	GHz	room	no (Poisson)
SPDC, heralded (Week 8)	10^{-2} (best 3×10^{-4})	0.1–4 MHz	room	heralded
Quantum dot in cavity	$\sim 10^{-3}$	tens of MHz	4–10 K	near-det.
NV center in diamond	0.1–0.3	10^4–10^6 s^{-1}	room	det., poor collection

Table 1: Order-of-magnitude comparison of single-photon sources; figures verified against the literature as of 2026-09 (cf. Fox Ch. 5; [3, 4]; NV row: [5, 6]).

The quantum-dot route is where cavity QED (Sec. 3) pays off: a cavity both speeds up emission (Purcell effect) and funnels the photons into one collectible mode.

Summary: Section 2

Click statistics classify light; $g^{(2)}(0) = 1 + [\sigma^2(n) - \langle n\rangle]/\langle n\rangle^2$ is the sharp tool: thermal 2, coherent 1, single photon 0. Classically $g^{(2)}(0) \geq 1$; antibunching is the fingerprint of the photon (1977). A dim laser is never a single-photon source. Real sources trade off purity, rate, temperature, and determinism.

3 Cavity QED: Redesigning the Vacuum

3.1 The Jaynes–Cummings model in one page

Put a two-level atom ($|E_v\rangle$,$|E_c\rangle$, splitting $\hbar\omega_{cv}$) in a cavity supporting a single mode of frequency ω. With the rotating-wave approximation, light–matter interaction reduces to the Jaynes–Cummings Hamiltonian (QO_note_4, Sec. 4)

$$\mathcal{H}_{\mathrm{JC}} = \hbar\omega\, a^\dagger a + \frac{\hbar\omega_{cv}}{2}\sigma_z + \hbar\lambda\left(\sigma_+ a + \sigma_- a^\dagger\right), \tag{3.1}$$

where λ is the coupling rate set by the dipole moment and the mode's vacuum field. Because the excitation number is conserved, the dynamics splits into 2×2 blocks $\{|E_c\rangle|n\rangle,\ |E_v\rangle|n+1\rangle\}$: the atom and the cavity coherently swap one quantum at the Rabi frequency $2\lambda\sqrt{n+1}$ (on resonance).[2] The eigenstates — *dressed states* — are split by $2\hbar\lambda\sqrt{n+1}$.

The striking case is $n=0$: even with ***no photon in the cavity***, the excited atom oscillates, $|E_c\rangle|0\rangle \leftrightarrow |E_v\rangle|1\rangle$ — the ***vacuum Rabi oscillation*** (Fig. 4). The "empty" cavity is not empty; its zero-point field drives the atom. (With a coherent drive the oscillations collapse and revive — see QO_note_4, Sec. 4.3, for this beautiful signature of field quantization.)

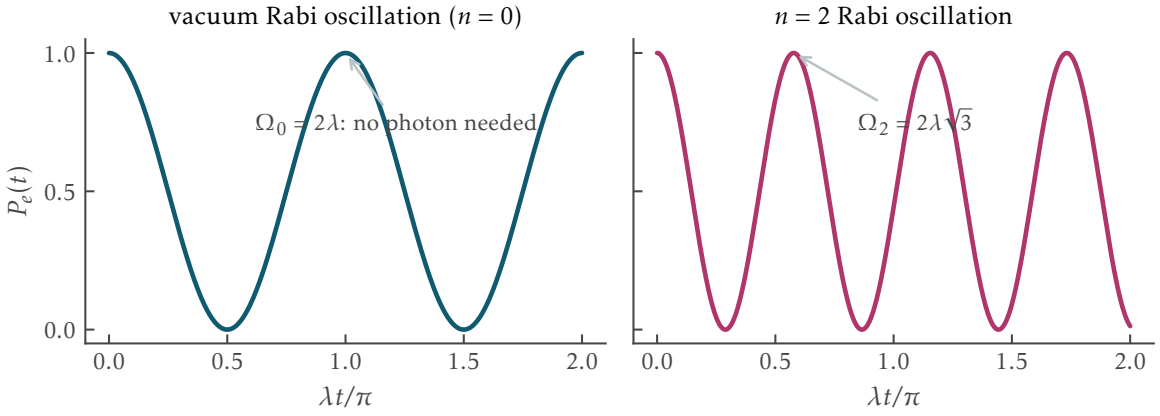


Figure 4: Jaynes–Cummings Rabi oscillations, $P_e(t) = \cos^2\left(\lambda\sqrt{n+1}\,t\right)$, for $n=0$ (vacuum) and $n=2$; the oscillation persists at $n=0$.

3.2 Spontaneous emission and the Purcell effect

A single lossless mode gives *reversible* exchange — the strong-coupling regime, $2\lambda > \kappa$, with $\kappa = \omega/Q$ the cavity decay rate. A leaky cavity, or a continuum of modes, gives *irreversible* decay at a rate set by the local density of modes — the weak-coupling, or Purcell, regime, $2\lambda < \kappa$. In JC language $F_P = 4\lambda^2/(\kappa\gamma)$, with γ the free-space rate; since $\lambda^2 \propto 1/V$ and $1/\kappa \propto Q$, this is the Q/V of Eq. (3.3) below.

Fermi's golden rule says the emission rate of an excited atom is proportional to the density of final photon states at its transition frequency (QO_note_4, Sec. 2.2):

$$W = \frac{\pi}{2\hbar^2}\left|\langle c|\mathcal{H}_I|v\rangle\right|^2 \rho(\omega) \implies W_{\mathrm{free}} = \frac{\omega^3 |d_{cv}|^2}{3\pi\epsilon_0\hbar c^3}, \tag{3.2}$$

[2] QO_note_4 calls $\omega_+ = \lambda\sqrt{n+1}$ the Rabi frequency (of the amplitude); the population oscillates at $2\omega_+ = 2\lambda\sqrt{n+1}$, the Rabi frequency Ω_n in Gerry & Knight's convention used here.

where the free-space result uses the vacuum density of modes $g(\omega) = \omega^2/\pi^2 c^3$ (QO_note_2, Sec. 3.1).[3] Read this formula the other way around: ***spontaneous emission is not a property of the atom alone.*** It is stimulated by the vacuum fluctuations of the modes around the atom — so if you change the modes, you change the rate. A cavity of quality factor Q and mode volume V concentrates the vacuum field at its resonance; the emission rate is enhanced by the Purcell factor (Purcell, 1946 [2]; result only)

$$F_P = \frac{W_{\text{cav}}}{W_{\text{free}}} = \frac{3}{4\pi^2}\left(\frac{\lambda_0}{n}\right)^3 \frac{Q}{V}, \tag{3.3}$$

for an emitter on resonance and at the field maximum. Detuned from the cavity, emission is *suppressed* instead. **Change the structure of the vacuum, and you change the atom's fate.** Keep Eq. (3.3) in mind: in Week 10 we will see that nanophotonics is precisely the engineering of Q/V — of the vacuum itself — and that this turns quantum-sensing limits into design parameters.

In-class discussion 3. *An atom is placed in a resonant cavity. Which of its properties change?* — Its emission rate (faster on resonance, slower off resonance), its emission direction (into the cavity mode), and even its energy levels (dressed-state splitting). The vacuum has been redesigned around it.

3.3 Detecting one photon: the SNSPD

Certifying $g^{(2)}(0) \approx 0$ requires detectors that click on a single photon. Photomultipliers (PMTs) and avalanche photodiodes (SPADs) do this with quantum efficiencies of order 10–70% and nanosecond timing. The current workhorse of quantum optics is the *superconducting nanowire single-photon detector* (SNSPD): a ~ 100-nm-wide superconducting wire (NbN, NbTiN, or WSi), cooled below its critical temperature (~1–4 K) and biased just below its critical current. One absorbed photon creates a local hotspot, the wire momentarily turns resistive, and a voltage pulse marks the arrival time. State-of-the-art SNSPDs combine system detection efficiency above 90% — $98.0 \pm 0.5\%$ at 1550 nm [7], and 99.5% claimed at 1350 nm [8] — with dark-count rates that can be pushed below 1 Hz (milli-Hz demonstrated) and timing jitter down to 2.6–4.3 ps [9]. Every experiment in Weeks 8–9 — QKD, Bell tests, boson sampling — stands on these detectors.

3.4 Coda: Glauber's photon

Roy Glauber received the 2005 Nobel Prize for the quantum theory of optical coherence — the theory behind $g^{(1)}$, $g^{(2)}$, and the coherent states used throughout this week. His central insight is the one we started from: light is described by quantum states of *modes*, and a photon is nothing more (and nothing less) than a single excitation of one of them. **Modes first; photons second.** Next week we take two photons at a time and let them interfere.

Summary: Section 3

JC physics: atom + cavity mode swap one quantum at rate $2\lambda\sqrt{n+1}$; vacuum Rabi oscillations exist at $n = 0$. Spontaneous emission $\propto$ density of vacuum modes $\Rightarrow$ Purcell factor

[3] The textbook form $(2\pi/\hbar)|M|^2\rho(E)$ differs by the factor-4 convention of QO_note_4, whose drive is written as $V_0 \cos\omega t$ (so each rotating term carries $V_0/2$).

$F_P = \frac{3}{4\pi^2}(\lambda_0/n)^3 Q/V$: the vacuum is designable. SNSPDs (superconducting nanowires) detect single photons with $> 90\%$ efficiency and ps jitter, enabling all the quantum optics that follows.

References

[1] H. J. Kimble, M. Dagenais, and L. Mandel, *Photon antibunching in resonance fluorescence*, Phys. Rev. Lett. **39**, 691 (1977).

[2] E. M. Purcell, *Spontaneous emission probabilities at radio frequencies*, Phys. Rev. **69**, 681 (1946).

[3] M. D. Eisaman, J. Fan, A. Migdall, and S. V. Polyakov, *Invited Review: Single-photon sources and detectors*, Rev. Sci. Instrum. **82**, 071101 (2011).

[4] P. Senellart, G. Solomon, and A. White, *High-performance semiconductor quantum-dot single-photon sources*, Nat. Nanotechnol. **12**, 1026 (2017).

[5] C. Kurtsiefer, S. Mayer, P. Zarda, and H. Weinfurter, *Stable solid-state source of single photons*, Phys. Rev. Lett. **85**, 290 (2000).

[6] I. Aharonovich, D. Englund, and M. Toth, *Solid-state single-photon emitters*, Nat. Photonics **10**, 631 (2016).

[7] D. V. Reddy, R. R. Nerem, S. W. Nam, R. P. Mirin, and V. B. Verma, *Superconducting nanowire single-photon detectors with 98% system detection efficiency at 1550 nm*, Optica **7**, 1649 (2020).

[8] J. Chang *et al.*, *Detecting telecom single photons with* $99.5^{+0.5}_{-2.07}\%$ *system detection efficiency and high time resolution*, APL Photonics **6**, 036114 (2021).

[9] B. Korzh *et al.*, *Demonstration of sub-3 ps temporal resolution with a superconducting nanowire single-photon detector*, Nat. Photonics **14**, 250 (2020).

[10] M. Fox, *Quantum Optics: An Introduction*, Oxford University Press (2006), Ch. 5–7 and 10.

[11] C. Gerry and P. Knight, *Introductory Quantum Optics*, Cambridge University Press (2005).